Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Silva Brothers Dairy #2

Physical address of dairy:

Number and Street	City	County	Zip Code
25723 N Elliott RD	Galt	San Joaquin	95632

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X007-X130-XX20-XXXX

B. OPERATORS

Silva, Joe

Operator name: Silva, Joe

Telephone no.: (209) 334-2914 Landline, (209) 607-7197 Cellular

Mailing Address Number and Street	City	State	Zip Code
12997 E Peltier RD	Acampo	CA	95220

This operator is responsible for paying permit fees.

C. OWNERS

Rausser, Gary

Legal owner name: Rausser, Gary

Telephone no.: (209) 368-0654 Landline, Cellular

Mailing Address Number and Street	City	State	Zip Code
12000 E Liberty RD	Galt	CA	95632

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	170	10	0	0	0	0
Maximum number	170	10	0	0	0	0
Average number	170	10	0	0	0	0
Avg live weight (lbs)	1,300	1,350	0	0		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 64 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 4,382.41 *tons per reporting period*

Total nitrogen from manure: 57,274.47 *lbs per reporting period*

After ammonia losses (30% loss applied): 40,092.13 *lbs per reporting period*

Total phosphorus from manure: 9,605.34 *lbs per reporting period*

Total potassium from manure: 31,886.43 *lbs per reporting period*

Total salt from manure: 82,344.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 4,612,500 *gallons*

Total nitrogen generated: 21,860.25 *lbs*

Total phosphorus generated: 1,946.28 *lbs*

Total potassium generated: 34,701.64 *lbs*

Total salt generated: 261,064.98 *lbs*

	4,612,500	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	4,612,500	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW #1	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	Corral solids *Separator Manure*	800.00 *ton*	Dry-weight	70.5	27,600.00	8,440.00	26,300.00		32.90

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	13,027.20	3,983.68	12,413.60	155,288.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	13,027.20	3,983.68	12,413.60	155,288.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	4	4	1	process wastewater	X007-X130-XX18-XXXX
Field 2	14	14	2	both	X007-X130-XX18-XXXX X007-X130-XX19-XXXX
Field 3	4	4	2	both	X007-X130-XX20-XXXX
Field 4	26	26	2	both	X007-X130-XX21-XXXX X007-X130-XX22-XXXX
Field 5	10	10	2	manure	X007-X130-XX21-XXXX X007-X130-XX22-XXXX
Field 6	25	25	2	both	X007-X130-XX22-XXXX
Field 7	32	32	2	both	X007-X130-XX22-XXXX
Totals for areas that were used for application	115	115	13		
Totals for areas that were not used for application					
Land application area totals	115	115	13		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

01/01/2023: Pasture

Crop: Pasture Acres planted: 4 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	60.00 *ton*	As-is		70.0	9,180.00	1,270.00	11,100.00		14.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	276.00	39.00	283.50	0.00
Total actual harvest content	15.00	275.40	38.10	333.00	1,287.00

Field 2

Field name: Field 2

Field 2

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 14 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	242.20 *ton*	As-is		68.2	3,240.00	630.00	5,460.00		9.04

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	17.30	112.10	21.80	188.92	994.65

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	382.60 *ton*	As-is		73.4	3,170.00	710.00	3,760.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	27.33	173.26	38.81	205.51	1,119.49

Field 3

Field name: Field 3

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 4 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	69.20 *ton*	As-is		68.2	3,240.00	630.00	5,460.00		9.04

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	17.30	112.10	21.80	188.92	994.65

Field 3

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 4 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	109.20 *ton*	As-is		73.4	3,170.00	710.00	3,760.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	27.30	173.08	38.77	205.30	1,118.32

Field 4

Field name: Field 4

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 26 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	445.00 *ton*	As-is		73.0	3,290.00	530.00	4,950.00		8.28

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	17.12	112.62	18.14	169.44	765.26

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	686.40 *ton*	As-is		67.4	3,980.00	610.00	5,280.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.40	210.14	32.21	278.78	1,170.47

Field 5

Field name: Field 5

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	176.80 *ton*	As-is		73.0	3,290.00	530.00	4,950.00		8.28

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	17.68	116.33	18.74	175.03	790.51

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	255.10 *ton*	As-is		67.4	3,980.00	610.00	5,280.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.51	203.06	31.12	269.39	1,131.01

Field 6

Field name: Field 6

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	404.20 *ton*	As-is		71.4	5,610.00	1,020.00	8,110.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.17	181.40	32.98	262.24	1,091.28

Field 6

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	626.30 *ton*	As-is		61.8	3,510.00	1,030.00	2,410.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.05	175.87	51.61	120.75	1,090.96

Field 7

Field name: Field 7

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 32 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	550.00 *ton*	As-is		66.6	3,740.00	700.00	5,920.00		0.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	17.19	128.56	24.06	203.50	22.96

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 32 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	833.70 *ton*	As-is		63.2	3,160.00	930.00	3,650.00		6.36

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.05	164.66	48.46	190.19	1,219.54

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 01/01/2023: Pasture

Field name: Field 1

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	170.86	29.67	174.93	3,223.26	150,000.00 *gal*
IW #1	Ground water	1.35	0.00	0.00	91.98	264,000.00 *gal*
Application event totals		172.21	29.67	174.93	3,315.23	

05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.02	0.00	0.00	137.97	396,000.00 *gal*
Application event totals		2.02	0.00	0.00	137.97	

05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.02	0.00	0.00	137.97	396,000.00 *gal*
Application event totals		2.02	0.00	0.00	137.97	

06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.02	0.00	0.00	137.97	396,000.00 *gal*
Application event totals		2.02	0.00	0.00	137.97	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.02	0.00	0.00	137.97	396,000.00 *gal*
Application event totals		2.02	0.00	0.00	137.97	

Field 1 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.02	0.00	0.00	137.97	396,000.00 *gal*
Application event totals		2.02	0.00	0.00	137.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.02	0.00	0.00	137.97	396,000.00 *gal*
Application event totals		2.02	0.00	0.00	137.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.02	0.00	0.00	137.97	396,000.00 *gal*
Application event totals		2.02	0.00	0.00	137.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.02	0.00	0.00	137.97	396,000.00 *gal*
Application event totals		2.02	0.00	0.00	137.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.02	0.00	0.00	137.97	396,000.00 *gal*
Application event totals		2.02	0.00	0.00	137.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	103.27	16.21	118.29	863.71	150,000.00 *gal*
IW #1	Ground water	1.35	0.00	0.00	91.98	264,000.00 *gal*
Application event totals		104.62	16.21	118.29	955.69	

Field 2 - 10/31/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Manure	Separator solids	116.31	35.57	110.84	1,386.50	100.00 *ton*
Application event totals		116.31	35.57	110.84	1,386.50	

01/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.19	3.34	68.06	887.40	187,500.00 *gal*
Application event totals		32.19	3.34	68.06	887.40	

Field 2 - 06/13/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.70	0.00	0.00	183.96	1,848,000.00 *gal*
Application event totals		2.70	0.00	0.00	183.96	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	194.69	15.29	281.64	1,260.69	375,000.00 *gal*
IW #1	Ground water	2.02	0.00	0.00	137.97	1,386,000.00 *gal*
Application event totals		196.72	15.29	281.64	1,398.66	

Field 2 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.60	0.00	0.00	177.39	1,782,000.00 *gal*
Application event totals		2.60	0.00	0.00	177.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.60	0.00	0.00	177.39	1,782,000.00 *gal*
Application event totals		2.60	0.00	0.00	177.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.60	0.00	0.00	177.39	1,782,000.00 *gal*
Application event totals		2.60	0.00	0.00	177.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.60	0.00	0.00	177.39	1,782,000.00 *gal*
Application event totals		2.60	0.00	0.00	177.39	

Field 3 - 10/31/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Manure	Separator solids	81.42	24.90	77.59	970.55	20.00 *ton*
Application event totals		81.42	24.90	77.59	970.55	

Field 3 - 10/31/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.59	7.02	142.93	1,863.54	112,500.00 *gal*
Application event totals		67.59	7.02	142.93	1,863.54	

Field 3 - 06/12/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.70	0.00	0.00	183.96	528,000.00 *gal*
Application event totals		2.70	0.00	0.00	183.96	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	204.43	16.05	295.73	1,323.73	112,500.00 *gal*
IW #1	Ground water	2.02	0.00	0.00	137.97	396,000.00 *gal*
Application event totals		206.45	16.05	295.73	1,461.69	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.70	0.00	0.00	183.96	528,000.00 *gal*
Application event totals		2.70	0.00	0.00	183.96	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.70	0.00	0.00	183.96	528,000.00 *gal*
Application event totals		2.70	0.00	0.00	183.96	

Field 3 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.53	0.00	0.00	172.46	495,000.00 *gal*
Application event totals		2.53	0.00	0.00	172.46	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.53	0.00	0.00	172.46	495,000.00 *gal*
Application event totals		2.53	0.00	0.00	172.46	

Field 4 - 10/31/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	112.74	34.47	107.43	1,343.84	180.00 *ton*
Application event totals		112.74	34.47	107.43	1,343.84	

02/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.66	3.60	73.30	955.66	375,000.00 *gal*
Application event totals		34.66	3.60	73.30	955.66	

Field 4 - 06/12/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage Plant date: 06/12/2023

Field 4 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	233.21	107.12	244.58	5,308.80	260.00 *ton*
Application event totals		233.21	107.12	244.58	5,308.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.70	0.00	0.00	183.96	3,432,000.00 *gal*
Application event totals		2.70	0.00	0.00	183.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.59	0.00	0.00	176.88	3,300,000.00 *gal*
Application event totals		2.59	0.00	0.00	176.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.59	0.00	0.00	176.88	3,300,000.00 *gal*
Application event totals		2.59	0.00	0.00	176.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.59	0.00	0.00	176.88	3,300,000.00 *gal*
Application event totals		2.59	0.00	0.00	176.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.59	0.00	0.00	176.88	3,300,000.00 *gal*
Application event totals		2.59	0.00	0.00	176.88	

Field 4 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.49	0.00	0.00	169.81	3,168,000.00 *gal*
Application event totals		2.49	0.00	0.00	169.81	

Field 5 - 10/31/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Manure	Separator solids	130.27	39.84	124.14	1,552.88	80.00 *ton*
Application event totals		130.27	39.84	124.14	1,552.88	

Field 5 - 06/11/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	233.21	107.12	244.58	5,308.80	100.00 *ton*
Application event totals		233.21	107.12	244.58	5,308.80	

Field 5 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.83	0.00	0.00	193.16	1,386,000.00 *gal*
Application event totals		2.83	0.00	0.00	193.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.70	0.00	0.00	183.96	1,320,000.00 *gal*
Application event totals		2.70	0.00	0.00	183.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.56	0.00	0.00	174.76	1,254,000.00 *gal*
Application event totals		2.56	0.00	0.00	174.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.56	0.00	0.00	174.76	1,254,000.00 *gal*
Application event totals		2.56	0.00	0.00	174.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.56	0.00	0.00	174.76	1,254,000.00 *gal*
Application event totals		2.56	0.00	0.00	174.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.56	0.00	0.00	174.76	1,254,000.00 *gal*
Application event totals		2.56	0.00	0.00	174.76	

Field 6 - 11/01/2022: Oats, silage-soft dough

Field 6 - 11/01/2022: Oats, silage-soft dough

Field name: Field 6

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Manure	Separator solids	130.27	39.84	124.14	1,552.88	200.00 *ton*
Application event totals		130.27	39.84	124.14	1,552.88	

02/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.10	7.49	152.46	1,987.78	750,000.00 *gal*
Application event totals		72.10	7.49	152.46	1,987.78	

Field 6 - 06/10/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.81	0.00	0.00	191.32	3,432,000.00 *gal*
Application event totals		2.81	0.00	0.00	191.32	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.70	0.00	0.00	183.96	3,300,000.00 *gal*
Application event totals		2.70	0.00	0.00	183.96	

Field 6 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.59	0.00	0.00	176.60	3,168,000.00 *gal*
Application event totals		2.59	0.00	0.00	176.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.59	0.00	0.00	176.60	3,168,000.00 *gal*
Application event totals		2.59	0.00	0.00	176.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	218.05	17.12	315.44	1,411.97	750,000.00 *gal*
IW #1	Ground water	1.89	0.00	0.00	128.77	2,310,000.00 *gal*
Application event totals		219.94	17.12	315.44	1,540.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.59	0.00	0.00	176.60	3,168,000.00 *gal*
Application event totals		2.59	0.00	0.00	176.60	

Field 7 - 11/01/2022: Oats, silage-soft dough

Field name: Field 7

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Manure	Separator solids	122.13	37.35	116.38	1,455.83	240.00 *ton*
Application event totals		122.13	37.35	116.38	1,455.83	

Field 7 - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.33	5.85	119.11	1,552.95	750,000.00 *gal*
Application event totals		56.33	5.85	119.11	1,552.95	

Field 7 - 06/09/2023: Corn, silage

Field name: Field 7

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.74	0.00	0.00	186.83	4,290,000.00 *gal*
Application event totals		2.74	0.00	0.00	186.83	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.70	0.00	0.00	183.96	4,224,000.00 *gal*
Application event totals		2.70	0.00	0.00	183.96	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.61	0.00	0.00	178.21	4,092,000.00 *gal*
Application event totals		2.61	0.00	0.00	178.21	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.61	0.00	0.00	178.21	4,092,000.00 *gal*
Application event totals		2.61	0.00	0.00	178.21	

Field 7 - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	204.43	16.05	295.73	1,323.73	900,000.00 *gal*
IW #1	Ground water	2.02	0.00	0.00	137.97	3,168,000.00 *gal*
Application event totals		206.45	16.05	295.73	1,461.69	

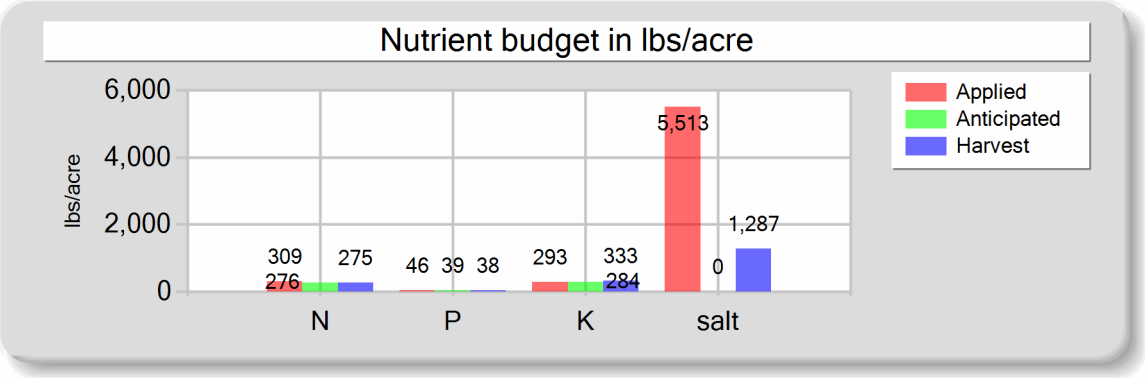
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.61	0.00	0.00	178.21	4,092,000.00 *gal*
Application event totals		2.61	0.00	0.00	178.21	

B. NUTRIENT BUDGET

Field 1 - 01/01/2023: Pasture

Field name: Field 1 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	274.13	45.88	293.22	4,086.96
Fresh water	20.92	0.00	0.00	1,425.67
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	309.05	45.88	293.22	5,512.63
Anticipated crop nutrient removal	276.00	39.00	283.50	0.00
Actual crop nutrient removal	275.40	38.10	333.00	1,287.00
Nutrient balance	33.65	7.78	-39.78	4,225.63
Applied to removed ratio	1.12	1.20	0.88	4.28

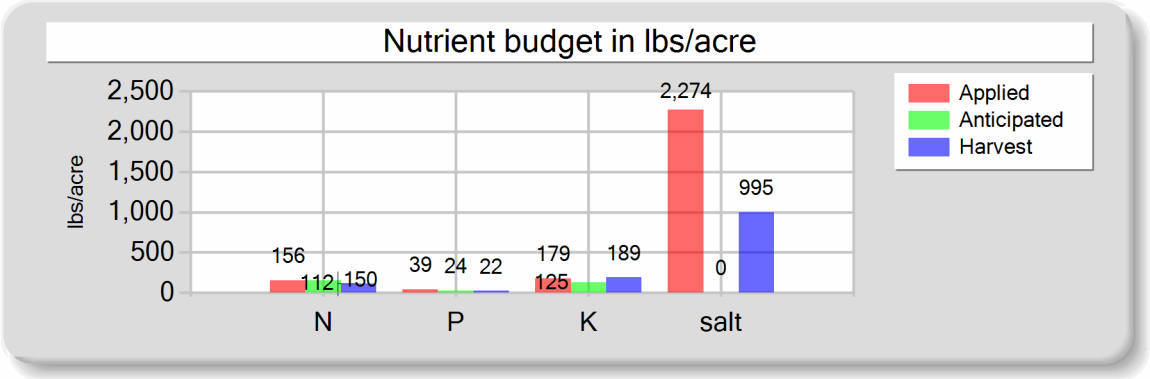
Fresh water applied	
4,092,000.00	*gallons*
150.69	*acre-inches*
37.67	*inches/acre*

Process wastewater applied	
300,000.00	*gallons*
11.05	*acre-inches*
2.76	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 10/31/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	116.31	35.57	110.84	1,386.50
Process wastewater	32.19	3.34	68.06	887.40
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	155.50	38.91	178.90	2,273.90
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	112.10	21.80	188.92	994.65
Nutrient balance	43.40	17.11	-10.02	1,279.25
Applied to removed ratio	1.39	1.79	0.95	2.29

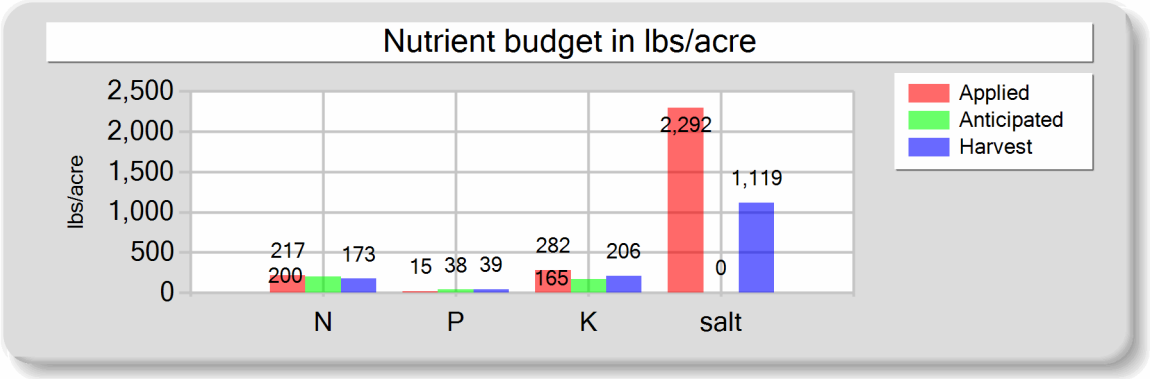
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
187,500.00 *gallons*
6.90 *acre-inches*
0.49 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 2 - 06/13/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	194.69	15.29	281.64	1,260.69
Fresh water	15.13	0.00	0.00	1,031.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	216.82	15.29	281.64	2,292.17
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	173.26	38.81	205.51	1,119.49
Nutrient balance	43.56	-23.52	76.13	1,172.68
Applied to removed ratio	1.25	0.39	1.37	2.05

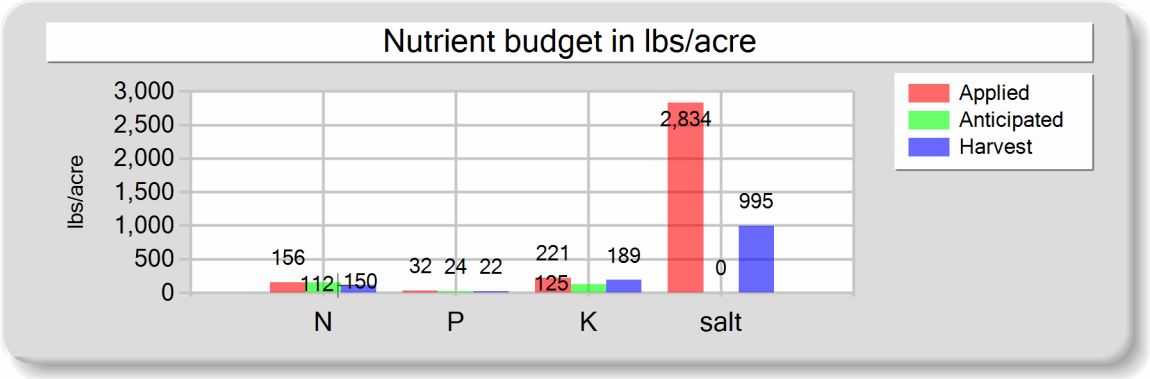
Fresh water applied	
10,362,000.00	*gallons*
381.60	*acre-inches*
27.26	*inches/acre*

Process wastewater applied	
375,000.00	*gallons*
13.81	*acre-inches*
0.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 10/31/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	81.42	24.90	77.59	970.55
Process wastewater	67.59	7.02	142.93	1,863.54
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	156.01	31.92	220.52	2,834.09
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	112.10	21.80	188.92	994.65
Nutrient balance	43.91	10.12	31.60	1,839.44
Applied to removed ratio	1.39	1.46	1.17	2.85

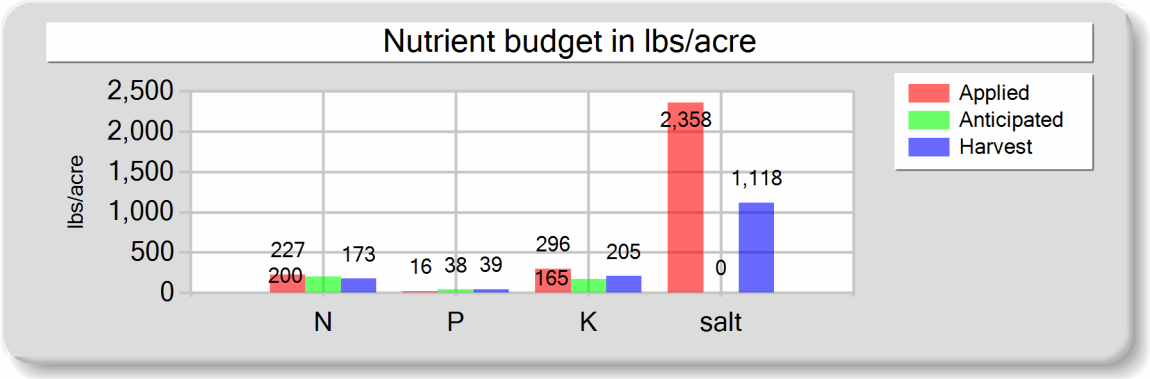
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
112,500.00 *gallons*
4.14 *acre-inches*
1.04 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 3 - 06/12/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	204.43	16.05	295.73	1,323.73
Fresh water	15.18	0.00	0.00	1,034.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	226.61	16.05	295.73	2,358.48
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	173.08	38.77	205.30	1,118.32
Nutrient balance	53.53	-22.71	90.43	1,240.17
Applied to removed ratio	1.31	0.41	1.44	2.11

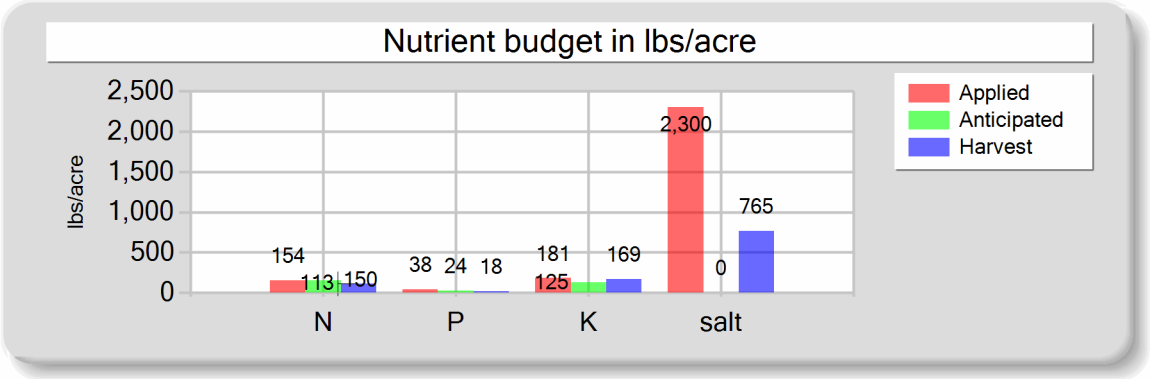
Fresh water applied	
2,970,000.00	*gallons*
109.38	*acre-inches*
27.34	*inches/acre*

Process wastewater applied	
112,500.00	*gallons*
4.14	*acre-inches*
1.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 10/31/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	112.74	34.47	107.43	1,343.84
Process wastewater	34.66	3.60	73.30	955.66
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	154.40	38.07	180.72	2,299.50
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	112.62	18.14	169.44	765.26
Nutrient balance	41.78	19.93	11.28	1,534.24
Applied to removed ratio	1.37	2.10	1.07	3.00

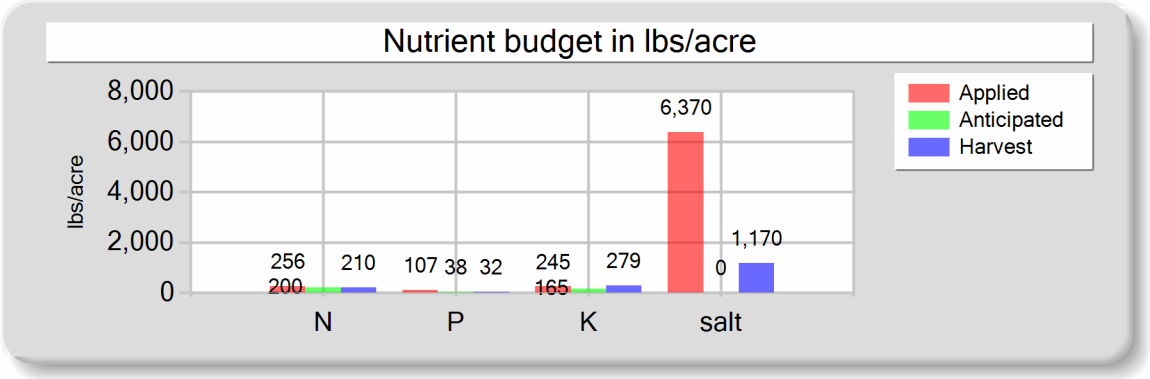
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
375,000.00	*gallons*
13.81	*acre-inches*
0.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 06/12/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	233.21	107.12	244.58	5,308.80
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	15.57	0.00	0.00	1,061.29
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	255.78	107.12	244.58	6,370.09
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	210.14	32.21	278.78	1,170.47
Nutrient balance	45.63	74.92	-34.20	5,199.62
Applied to removed ratio	1.22	3.33	0.88	5.44

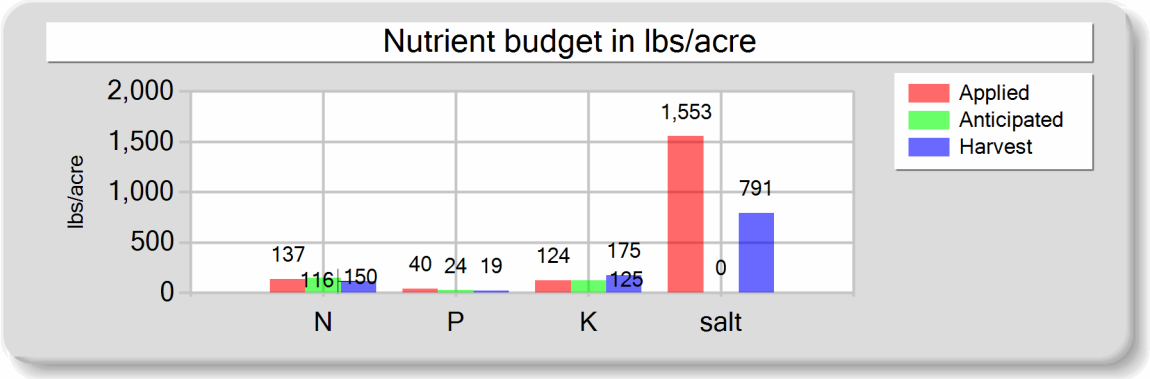
Fresh water applied	
19,800,000.00	*gallons*
729.17	*acre-inches*
28.04	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 10/31/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	130.27	39.84	124.14	1,552.88
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	137.27	39.84	124.14	1,552.88
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	116.33	18.74	175.03	790.51
Nutrient balance	20.94	21.10	-50.90	762.37
Applied to removed ratio	1.18	2.13	0.71	1.96

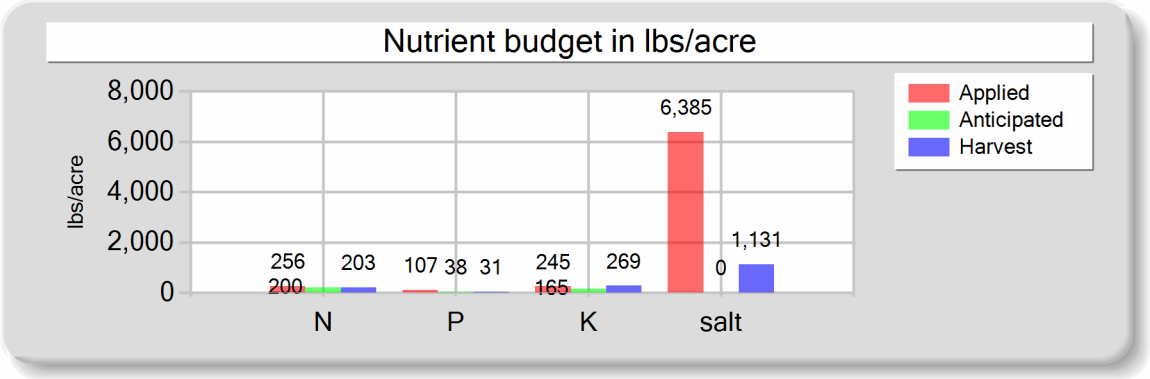
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 5 - 06/11/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	233.21	107.12	244.58	5,308.80
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	15.79	0.00	0.00	1,076.15
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	256.00	107.12	244.58	6,384.95
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	203.06	31.12	269.39	1,131.01
Nutrient balance	52.94	76.00	-24.80	5,253.94
Applied to removed ratio	1.26	3.44	0.91	5.65

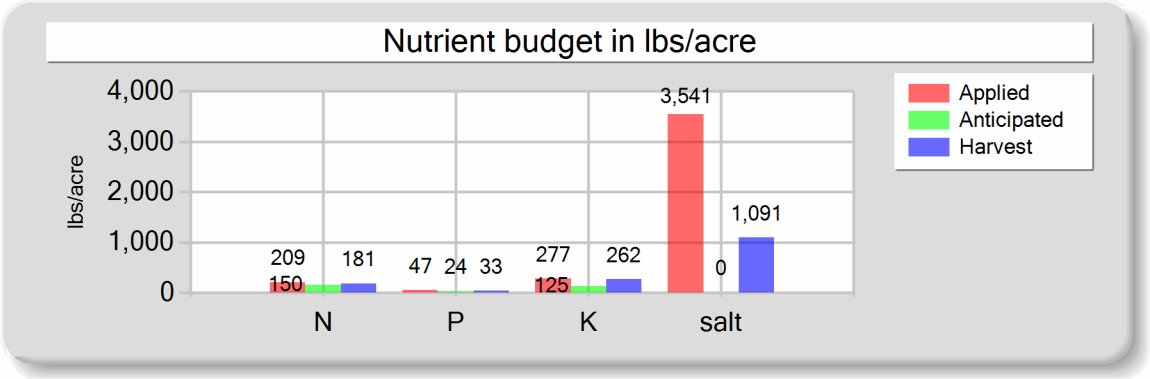
Fresh water applied	
7,722,000.00	*gallons*
284.38	*acre-inches*
28.44	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 11/01/2022: Oats, silage-soft dough

Field name: Field 6 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	130.27	39.84	124.14	1,552.88
Process wastewater	72.10	7.49	152.46	1,987.78
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.37	47.32	276.60	3,540.66
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	181.40	32.98	262.24	1,091.28
Nutrient balance	27.97	14.34	14.35	2,449.38
Applied to removed ratio	1.15	1.43	1.05	3.24

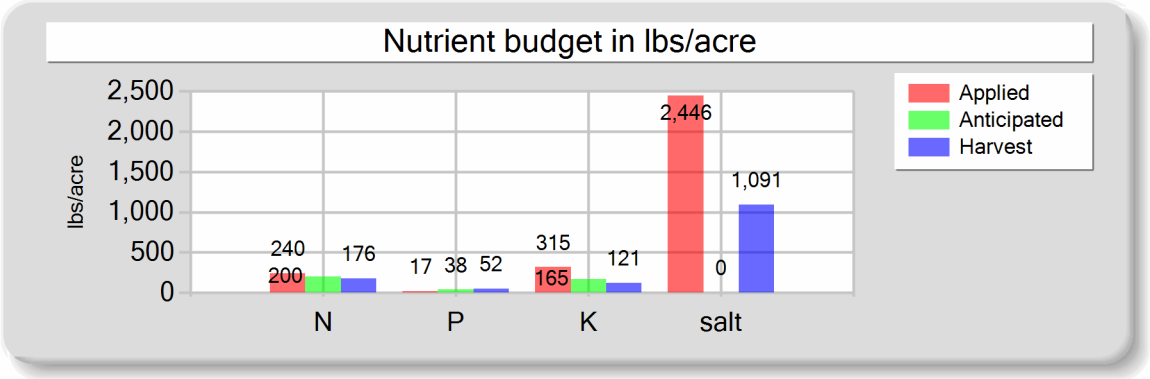
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
750,000.00 *gallons*
27.62 *acre-inches*
1.10 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 6 - 06/10/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	218.05	17.12	315.44	1,411.97
Fresh water	15.17	0.00	0.00	1,033.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.22	17.12	315.44	2,445.81
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	175.87	51.61	120.75	1,090.96
Nutrient balance	64.36	-34.48	194.69	1,354.85
Applied to removed ratio	1.37	0.33	2.61	2.24

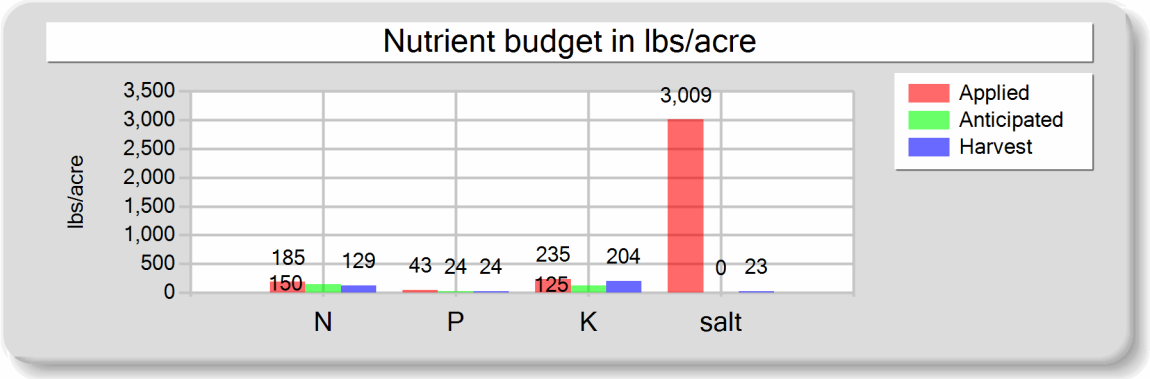
Fresh water applied	
18,546,000.00	*gallons*
682.99	*acre-inches*
27.32	*inches/acre*

Process wastewater applied	
750,000.00	*gallons*
27.62	*acre-inches*
1.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 11/01/2022: Oats, silage-soft dough

Field name: Field 7 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	122.13	37.35	116.38	1,455.83
Process wastewater	56.33	5.85	119.11	1,552.95
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	185.46	43.20	235.49	3,008.78
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	128.56	24.06	203.50	22.96
Nutrient balance	56.90	19.13	31.99	2,985.81
Applied to removed ratio	1.44	1.80	1.16	131.03

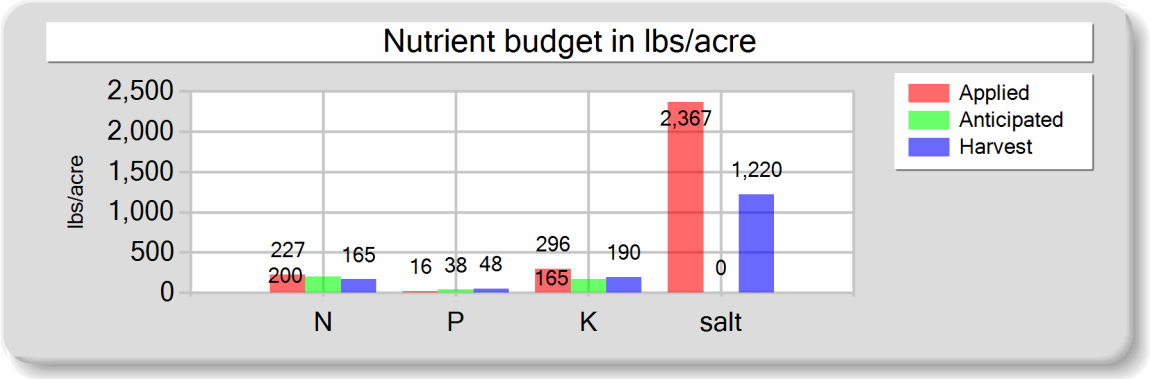
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
750,000.00 *gallons*
27.62 *acre-inches*
0.86 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 7 - 06/09/2023: Corn, silage

Field name: Field 7 Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	204.43	16.05	295.73	1,323.73
Fresh water	15.31	0.00	0.00	1,043.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	226.73	16.05	295.73	2,367.11
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	164.66	48.46	190.19	1,219.54
Nutrient balance	62.08	-32.41	105.54	1,147.57
Applied to removed ratio	1.38	0.33	1.55	1.94

Fresh water applied	
23,958,000.00	*gallons*
882.29	*acre-inches*
27.57	*inches/acre*

Process wastewater applied	
900,000.00	*gallons*
33.14	*acre-inches*
1.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Separator Manure imported from Silva Bros 1

Sample and source description: Separator Manure imported from Silva Bros 1

Sample date: 10/17/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,600.00	8,440.00	26,300.00							32.90
DL	100.00	100.00	100.00							0.01

Corral Manure

Sample and source description: Corral Manure

Sample date: 04/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,600.00	11,300.00	25,800.00	23,400.00	11,200.00	5,800.00	7,500.00	0.00		56.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Manure

Sample and source description: Corral Manure

Sample date: 10/10/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,300.00	8,210.00	25,100.00							27.70
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ1

Sample and source description: WWQ1

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	288.00	162.00	0.00	0.00	29.90	609.00	132.00	83.80	173.00	2,620.00	0.00	206.00	451.00	6,270.00	7,940
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 04/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	546.00	288.00	0.00	0.00	94.80	559.00								5,070.00	10,300
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	871.00	669.00	0.00	0.00	68.40	1,260.00								10,900.00	5,640
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	330.00	218.00	0.00	0.00	51.80	378.00								3,650.00	2,760
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

IW #1

IW #1

Sample description: IW #1

Sample date: 09/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.45	0.00	0.00								206.00	167
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

Field 1

Soil

Sample and source description: Soil

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			102.00				
DL			4.00				

Field 2

Soil

Sample and source description: Soil

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			102.00				
DL			4.00				

Field 3

Soil

Sample and source description: Soil

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			102.00				
DL			4.00				

Field 4

Soil

Sample and source description: Soil

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			36.00				
DL			4.00				

Field 5

Soil

Sample and source description: Soil

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			36.00				
DL			4.00				

Field 6

Field 6

Soil

Sample and source description: Soil

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			20.00				
DL			4.00				

Field 7

Soil

Sample and source description: Soil

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			38.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Field 1 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 84.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,740.00	660.00	5,710.00		14.30
DL	100.00	100.00	100.00		0.01

Field 2 - 10/31/2022: Oats, silage-soft dough

Field 2 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,240.00	630.00	5,460.00		9.04
DL	100.00	100.00	100.00		0.01

Field 2 - 06/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,170.00	710.00	3,760.00		7.70
DL	100.00	100.00	100.00		0.01

Field 3 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,240.00	630.00	5,460.00		9.04
DL	100.00	100.00	100.00		0.01

Field 3 - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,170.00	710.00	3,760.00		7.70
DL	100.00	100.00	100.00		0.01

Field 4 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,290.00	530.00	4,950.00		8.28
DL	100.00	100.00	100.00		0.01

Field 4 - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,980.00	610.00	5,280.00		6.80
DL	100.00	100.00	100.00		0.01

Field 5 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,290.00	530.00	4,950.00		8.28
DL	100.00	100.00	100.00		0.01

Field 5 - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,980.00	610.00	5,280.00		6.80
DL	100.00	100.00	100.00		0.01

Field 6 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,610.00	1,020.00	8,110.00		11.80
DL	100.00	100.00	100.00		0.01

Field 6 - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,510.00	1,030.00	2,410.00		5.70
DL	100.00	100.00	100.00		0.01

Field 7 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,740.00	700.00	5,920.00		0.20
DL	100.00	100.00	100.00		0.01

Field 7 - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,160.00	930.00	3,650.00		6.36
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

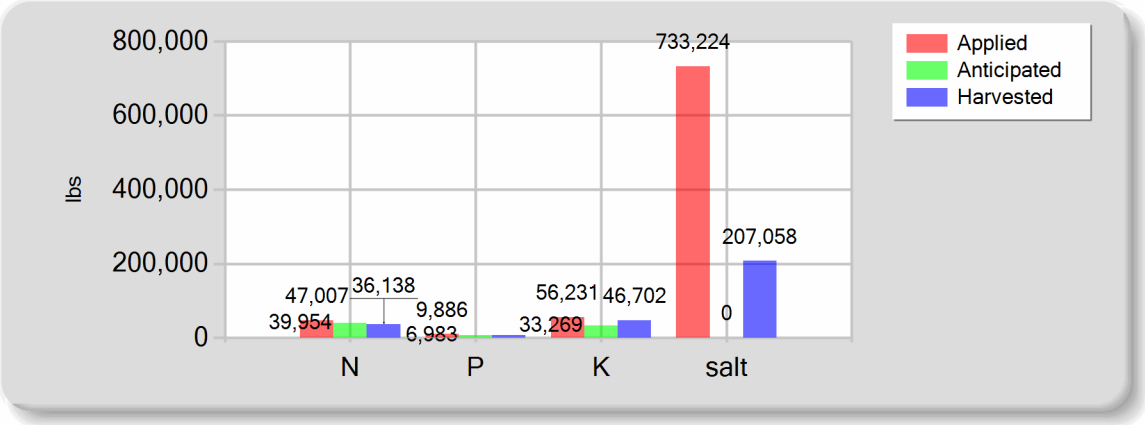
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

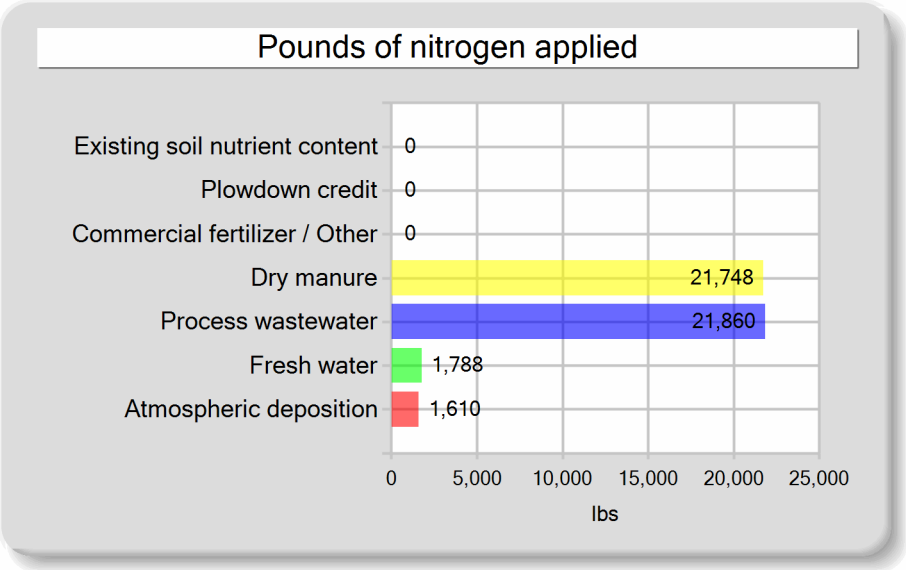
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

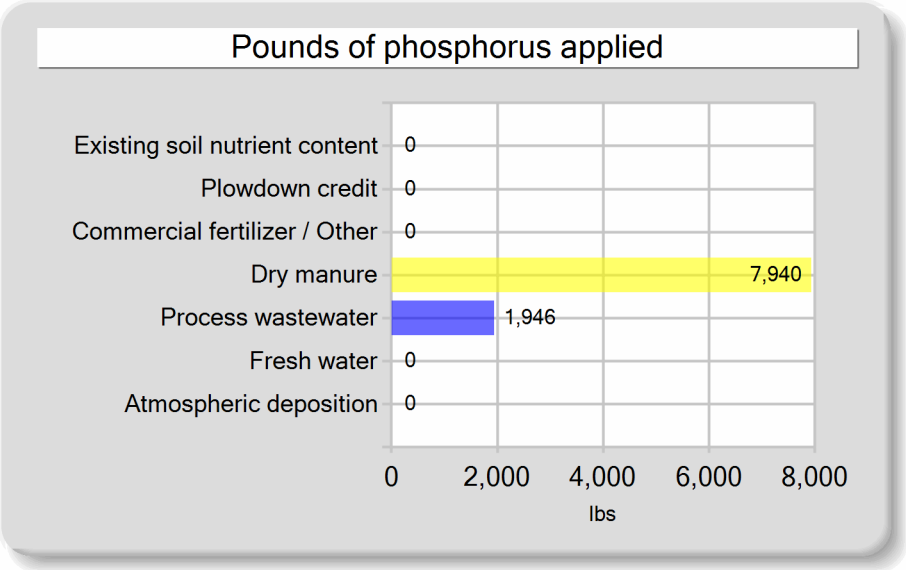
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	21,748.37	7,939.74	21,528.96	350,287.00
Process wastewater	21,860.25	1,946.28	34,701.64	261,064.98
Fresh water	1,787.94	0.00	0.00	121,871.63
Atmospheric deposition	1,610.00	0.00	0.00	0.00
Total nutrients applied	47,006.55	9,886.02	56,230.60	733,223.61
Anticipated crop nutrient removal	39,954.00	6,982.50	33,268.50	0.00
Actual crop nutrient removal	36,138.00	7,486.29	46,701.78	207,058.08
Nutrient balance	10,868.55	2,399.73	9,528.82	526,165.54
Applied to removed ratio	1.30	1.32	1.20	3.54

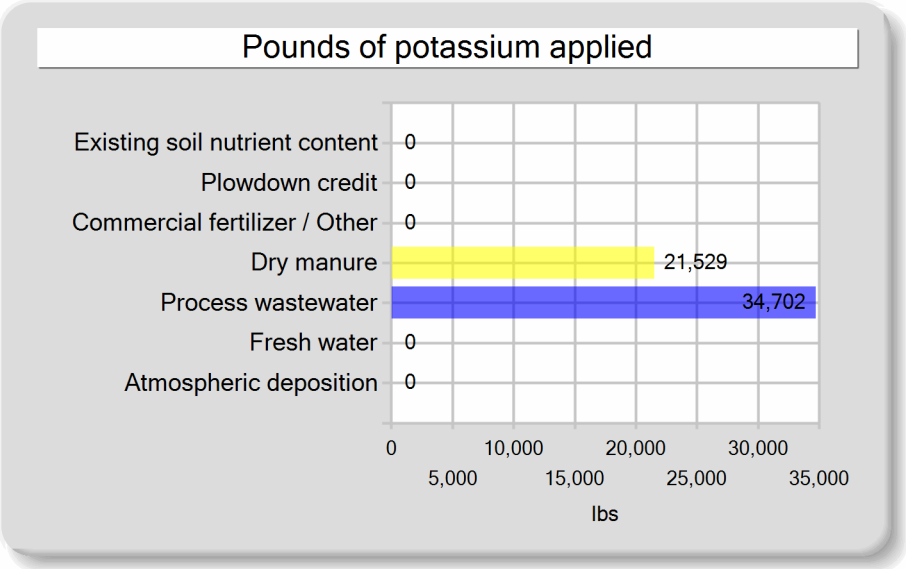
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

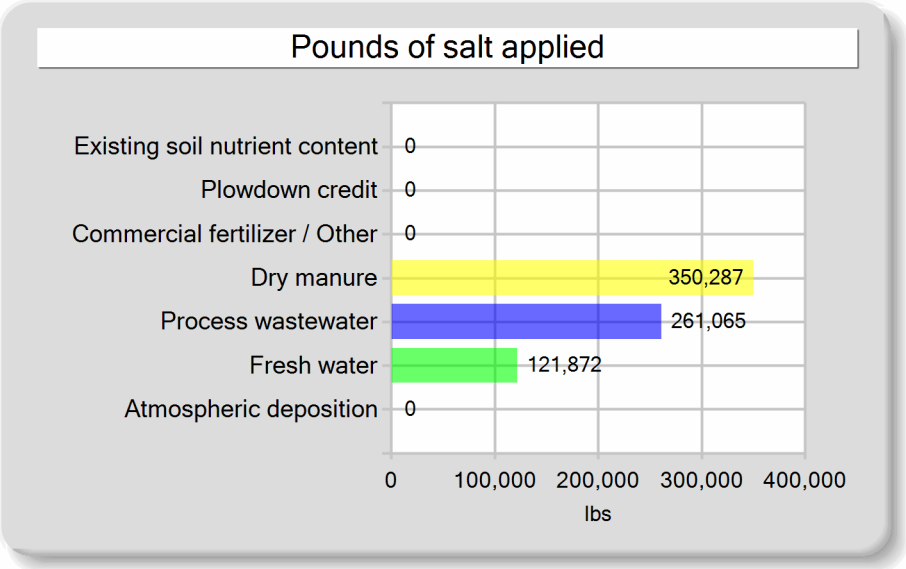


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field 7 Forage (1.44) had a high nitrogen field ratio as field failed to meet anticipated removal of 150 pounds of nitrogen/acre. The subject field removed only 128 pounds of nitrogen per acre. Yield for the subject field was adequate however a low nitrogen value from the plant tissue lab analysis resulted in 7.48 pounds of nitrogen removed per harvested ton compared to the anticipated 10 pounds of nitrogen removed per harvested ton ultimately resulting in the high nitrogen field ratio. Nitrogen application to the subject field was not considered excessive but in line with that recommended in the dairy's nutrient management plan.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Gary Rausser	Joe M Silva
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Gary Rausser	Joe Silva
PRINT OR TYPE NAME	PRINT OR TYPE NAME
4-26-24	4-26-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE™
LABORATORY INC

Silva Bros #2
12997 E Peltier Rd
Acampo, CA 95220

Account# 00-0026229
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 09/28/2023 15:15
Reported: 10/04/2023 12:29

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23I1833-01	DW Barn	Drinking Water			09/27/2023 7:00
23I1833-02	IW 1	Ag Water			09/27/2023 7:00

Default Cooler Temperature on Receipt °C: 12.4
Containers Intact
COC/Labels Agree

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

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LABORATORY INC

Silva Bros #2
12997 E Peltier Rd
Acampo, CA 95220

Account# 00-0026229
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 09/28/2023 15:15
Reported: 10/04/2023 12:29

Sample Results

Sample: DW Barn
23I1833-01 (Water)

Sampled: 9/27/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.20**	mmhos/cm	0.01	1		09/29/23 13:01	SM 2510 B		BEI1078
Electrical Conductivity umhos	**196**	umhos/cm	10.0	1		09/29/23 13:01	SM 2510 B		BEI1078
Ammonia (as N)	*	mg/L	0.00	1		09/27/23 07:00	Field		BEJ0131
Nitrate Nitrogen as NO3N	**2.0**	mg/L	0.1	1	10	09/29/23 01:46	EPA 300.0		BEI1030
Temperature	**25.0**	units	0.0	1		09/29/23 13:01	SM 4500-H+	H	BEI1078
pH	**7.6**	units	1.0	1		09/29/23 13:01	SM 4500-H+	H	BEI1078
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/03/23 11:08	SM 4500-NH3 C		BEJ0001
Total Nitrogen	**1.95**	mg/L	1.00	1		10/03/23 11:08	SM 4500-NH3 C		BEJ0001

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DELLAVALLE™
LABORATORY INC

Silva Bros #2
12997 E Peltier Rd
Acampo, CA 95220

Account# 00-0026229
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 09/28/2023 15:15
Reported: 10/04/2023 12:29

Sample Results
(Continued)

Sample: IW 1
23I1833-02 (Water)

Sampled: 9/27/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.21**	mmhos/cm	0.01	1		09/29/23 13:02	SM 2510 B		BEI1078
Electrical Conductivity umhos	**205**	umhos/cm	10.0	1		09/29/23 13:02	SM 2510 B		BEI1078
Nitrate Nitrogen as NO3N	**2.1**	mg/L	0.1	1	10	09/29/23 02:06	EPA 300.0		BEI1030
Temperature	**25.0**	units	0.0	1		09/29/23 13:02	SM 4500-H+	H	BEI1078
pH	**7.7**	units	1.0	1		09/29/23 13:02	SM 4500-H+	H	BEI1078
Total Filterable Solids (TDS)	**167**	mg/L	10.0	1		10/02/23 14:50	SM 2540 C		BEI1072
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/03/23 11:10	SM 4500-NH3 C		BEJ0001
Total Nitrogen	**2.45**	mg/L	1.00	1		10/03/23 11:10	SM 4500-NH3 C		BEJ0001

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WATER WORK REQUEST

26229

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

Purchase Order No

Bill To: 6 Acct # 13 Cons #

No. Samples: 2 No of Bottles: 4

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: Silva Bros #2
Address: 12997 E Pettier Rd
City: Acampo **State:** CA **Zip:** 95220
Telephone: **Fax:**
Cell/Email: 209 321-1318
COPY TO: F&A Ag Services
& JSawyer@Fragservices.com
REQUESTED BY: Jeanette Sawyer
PROJECT:
CROP:

Other:

Analysis and Bottles Required: (*Please indicate Analysis*)

DWW1 ++e
DCW1 ++e

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	DW Barn	9/27/23	Start 7 AM	12.4	
2	IW 1	"		7.3	
3					
9					
10					

09/28/23 15:15 23I1833

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second	JLF	DLV	9-28-23 10:15	9-28-23 2:35
Third				
Fourth	JLF	DLV	9/28/23 15:15	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature
Sample received in cooler with ice (coolant)
[ ] Yes [ ] No